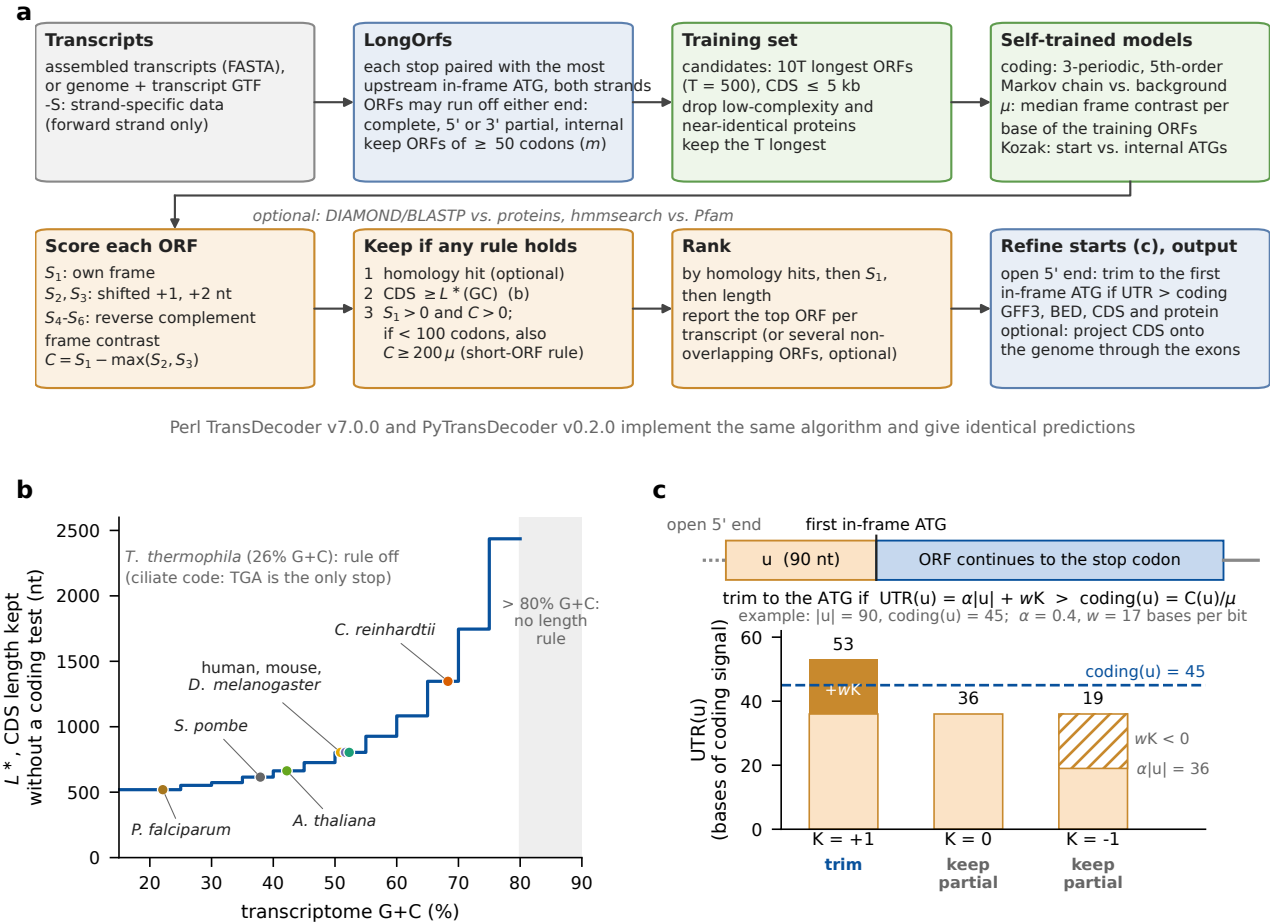
a
Transcripts
assembled transcripts (FASTA),
or genome + transcript GTF
-S: strand-specific data
(forward strand only)
LongOrfs
each stop paired with the most
upstream in-frame ATG, both strands
ORFs may run off either end:
complete, 5' or 3' partial, internal
keep ORFs of ≥ 50 codons (m)
Training set
candidates: 10T longest ORFs
(T = 500), CDS ≤ 5 kb
drop low-complexity and
near-identical proteins
keep the T longest
Self-trained models
coding: 3-periodic, 5th-order
Markov chain vs. background
μ: median frame contrast per
base of the training ORFs
Kozak: start vs. internal ATGs
optional: DIAMOND/BLASTP vs. proteins, hmmsearch vs. Pfam
Score each ORF
S1: own frame
S2, S3: shifted +1, +2 nt
S4-S6: reverse complement
frame contrast
C = S1 − max(S2, S3)
Keep if any rule holds
1 homology hit (optional)
2 CDS ≥ L* (GC) (b)
3 S1 > 0 and C > 0;
if < 100 codons, also
C ≥ 200 μ (short-ORF rule)
Rank
by homology hits, then S1,
then length
report the top ORF per
transcript (or several non-
overlapping ORFs, optional)
Refine starts (c), output
open 5' end: trim to the first
in-frame ATG if UTR > coding
GFF3, BED, CDS and protein
optional: project CDS onto
the genome through the exons
Perl TransDecoder v7.0.0 and PyTransDecoder v0.2.0 implement the same algorithm and give identical predictions
b
T. thermophila (26% G+C): rule off
(ciliate code: TGA is the only stop)
> 80% G+C:
no length
rule
C. reinhardtii
human, mouse,
D. melanogaster
S. pombe
P. falciparum
A. thaliana
L*, CDS length kept
without a coding test (nt)
transcriptome G+C (%)
c
open 5' end
first in-frame ATG
u (90 nt)
ORF continues to the stop codon
trim to the ATG if UTR(u) = α|u| + wK > coding(u) = C(u)/μ
example: |u| = 90, coding(u) = 45; α = 0.4, w = 17 bases per bit
UTR(u)
(bases of coding signal)
53
+wK
36
19
coding(u) = 45
wK < 0
α|u| = 36
K = +1
trim
K = 0
keep
partial
K = -1
keep
partial